

Intelligent Automatic Robotics Manipulator

(A landmine Detecting Robot)



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Abstract

Landmines and unexploded metallic objects remain a serious threat to human life, infrastructure and environmental safety particularly in post-conflict and remote regions. Conventional mine detection techniques are often slow, risky and highly dependent on human involvement which exposes personnel to extreme danger. To overcome these limitations, this project presents the design and development of an Intelligent Automatic Robotics Manipulator for landmine Detection and Retrieval using embedded intelligence and autonomous navigation. The proposed system is an ESP32-based mobile Robot integrated with a metallic Robotic Arm, GPS module and Metal Detection Sensor. It scans a predefined area, detects metallic objects and safely retrieval potential landmines without direct human intervention. The robot operates autonomously with reliable sensing and precise manipulation, offering a cost-effective and scalable solution for hazardous environment operation.

Keywords: *Landmine Detection, Robotics Manipulator, ESP32, GPS-Guided Navigation, Metal Detection, Autonomous Robotics, Hazardous Environment Automation.*

1 Introduction

1.1 Background

Land mines and other unexploded metallic materials from past conflicts continue to endanger people across the world. Even decades after conflicts have ended, these hidden devices remain buried in the ground, causing injuries, loss of life and long-term socio-economic problems. According to international humanitarian reports, thousands of civilians and soldiers are injured or killed every year due to accidental encounters with landmines especially border area and in rural and post-war regions where agricultural and development activities are essential for survival.

Conventional mine detection and clearance techniques largely depend on manual inspection using handheld metal detectors and trained personnel. Although these methods are widely used, they are extremely time-consuming, physically exhausting and most importantly, highly dangerous. Human operators are directly exposed to the risk of accidental detonation, harsh environmental conditions and limited visibility. As a result, there is a growing global demand for automated, intelligent and reliable system that can perform detection and handling tasks while keeping humans at a safe distance.

Advancements in embedded systems, robotics and sensor technologies have opened new opportunities for developing autonomous machines capable of operating in hazardous environments. Microcontrollers such as ESP32, combined with GPS module, metal detector sensors and robotics manipulators enable the creation of compact yet powerful robotics platforms. These technologies make it possible to design systems that can navigate, sense, decide and act with minimal human intervention. This project utilizes these modern tools to address a critical real-world safety challenge.

1.2 Problem Statement

- Despite ongoing demining efforts worldwide, existing methods still suffer from several major limitations:
- High risk to human life during detection and extraction processes.
- Slow and labor-intensive operations, especially in large or difficult terrains.
- Limited accuracy and consistency due to human fatigue and environmental interference.
- Lack of precise digital mapping and documentation of hazardous areas.
- Loss of life of civilian and soldiers a during critical days in hard area like border and war zone.

Manual mine detection not only endangers trained personnel but also increases the overall cost and duration of clearance operations. In many developing regions, access to advance equipment and skill experts is limited, making the problem even more critical.

Therefore, there is a strong need for an intelligent, low-cost and autonomous robotics system that can:

- Systematically scan a defined area
- Detect metallic objects and potential landmines
- Safely manipulate and retrieve detected items
- Provide accurate location information for monitoring and analysis.

This project aims to bridge the gap between safety, efficiency and affordability in landmine detection by introducing an automated robotics manipulator capable of performing these tasks with reduce human involvement.

1.3 Objectives of the Project

The main objectives of the project is to designed and develop an Intelligent Automatic Robotics Manipulator for landmine Detection and Retrieval using embedded and Robotics Technologies.

The specific objectives are:

- To design a mobile robotics platform controlled by an ESP32 microcontroller.
- To integrate a metal detection sensor for identifying buried or surface-level metallic objects.
- To implement a GPS-based area definition and navigation system for systematic scanning.
- To develop a robotics arm capable of precise and stable piking of detected objects.
- To enable autonomous operation with minimal human intervention.
- To ensure safe handling and reduce risk during detection and extraction process
- To record and monitor the geographical locations of detected hazards
- To create a cost-effective and scalable prototype suitable for future enhancement.

1.4 Scope of the project

This project's scope includes the design, development and implementation of Intelligent Automatic Robotics Manipulator in a war are for landmines Detection. The study covers the following aspects:

1.4.1 Mechanical and Electrical Design

Design and integration of a robotics manipulator and mobile platform using ESP32 microcontroller, motor driver and appropriate sensors.

1.4.2 Sensing and Detection Systems

Implementation of metal detection and proximity sensing techniques to identify potential landmine-like objects in the test environment.

1.4.3 Control and Navigation

Development of microcontroller-based control algorithm for movement, obstacle avoidance and manipulate operation.

1.4.4 Human-Robot Interaction

Basic user interface using monitoring and manual override to ensure safe and reliable operation

1.4.5 Experimental Validation

Testing the system in a controlled environment using simulated landmine object and evaluating detection accuracy and system reliability.

Although the primary focus of this project is landmine detection, the fundamental principles and design can be adapted for similar hazardous environments that require automated inspection and service operations.

1.5 Application

The Intelligent Automatic Robotics Manipulator has wide practical application in hazardous and security-sensitive environments. Its primary use is in humanitarian demining operations, where it can detect and safely remove landmines without exposing humans to danger. The system can be developed in post-conflict regions to clear agricultural land and residential area helping communities return to normal life.

It is also useful for military and defense operation for battlefield clearance and surveillance. The system can be applied in border security and restricted zones for automated inspection and threat detection. Additionally, it can be used in industrial safety for identifying and handling hazardous metallic objects in mining and construction sites.

The project also serves as a research platform for intelligent robotics, embedded systems and autonomous navigation studies, enabling future development in smart robotics technologies.

1.6 Significance of the study

The deployment of intelligent Automatic Robotics Manipulator is expected to provide numerous advantages:

- Reduce risk to human life by eliminating direct human involvement in landmine detection and handling.
- Provides a safer alternative to traditional manual demining method.
- Improve the speed and efficiency of hazardous area scanning and clearance.
- Enhance accuracy and consistency in detecting metallic threats.
- Enable precise GPS-based mapping of dangerous zones.
- Demonstrate practical integration of embedded systems and mechanical manipulation
- Contributes to the development of autonomous and intelligent robotic systems
- Serves as a foundation of future research in smart and safety-critical robotics.

Additionally, this study adds to the broader field of service robotics by showcasing the application of autonomous Robots for landmine detection and manipulation thereby broadening the potential focus on motion planning and manipulation of robot.

2. Literature Review

2.1 Introduction

Landmines are a persistent global threat, buried across many regions and causing civilian and soldier's injuries, deaths and socioeconomic disruption. Traditional landmine detection methods are slow, dangerous, risk of human life and costly motivating the development of robotics system that can detect and interact with buried explosive devices that can safely identify the landmine and pick from ground while keeping human at safe distance [1].

2.2 Challenges in Landmine Detection

Detecting landmines is inherently difficult due to variable soil conditions, heterogeneous environments and the small size and low signature of many mines. These issues often lead to high false positive rates in detecting systems, hindering reliability and efficiency [2]. For Example, magnetic and conductive soil properties such as those found in tropical or highly mineralized regions can mask or distort the electromagnetic response of sensors, reducing sensitivity and increasing false alarms from soil background noise and harmless metallic debris. This problem is particularly acute for modern anti-personnel mines that contain minimal metal which forces detectors to operate at high sensitivity yet still struggle to discriminate actual threats from soil effects or scattered scrap metal, further slowing detection and reducing operational effectiveness. Studies have shown that the physical and electrical characteristics of different soils directly influence detection performance and misclassification rates, making consistent identification across diverse terrains one of the biggest challenges in landmine clearance [3].

2.3 Conventional and Manual Demining Techniques

Manual probing, trained animals (e.g., Dogs), and handheld detectors are historically common tools for mine clearance. However, these methods expose operators to high risk, are time-consuming and often inefficient across large contaminated areas [4].

2.4 Robotics Systems for Mine Detection

The Concept of Robotic Demining has been under investigation for decades. Early work explored semi-autonomous robots to detect and mark mine locations, enabling safer clearance procedures [5]. Contemporary research expands this to intelligent robotics systems with advance sensing and autonomous capabilities [6].

2.5 Mobile Platforms and UGVs

The Unmanned Ground Vehicles (UGVs) provide crucial mobility for covering large fields and rugged terrains. Researchers have designed wheeled and track-based platform which equipped with navigation and sensing suites to systematically search minefield [7].

2.6 Sensor Integration and Detection Technology

Metal detectors remain fundamental sensors for identifying metallic components of mines. Several research prototypes integrate metal detectors with other sensor types such as ultrasonic sensor, infrared (IR) sensor and temperature sensor to improve detection and reliability [8].

2.7 GPS, GMS and IOT Integration

Modern Robotic systems are often combined with GPS positioning and GMS data links to determine accurate mine location and transmit detection results in real time. This provides better

operational insight and remote monitoring by operators. [9]. IoT-enhance designs further expand connectivity and data analysis capabilities [10].

2.8 Sensor Fusion and Multimodal Detection

Sensor fusion combining data from metal detectors, imaging systems and other sensing modalities has been shown to reduce false alarms and improve classification performance in mine detection. [11]. Leveraging multiple sensors can help differentiate mines from clutter and environmental noise.

2.9 Machine learning and AI Approaches

Recent advancements apply machine learning and computer vision algorithms to analyze sensor and image data from improved detection. For Example, researchers have developed IoT robotics that use real-time image classification to identify suspicious terrain features potentially indicating buried mines [12].

2.10 Date Sets and Algorithm Benchmarking

Benchmark datasets designed for humanitarian demining robotics, such as Mine Insight, provide multispectral and multi-view sensor data specially for algorithm evaluation. These datasets are crucial for training and validating AI systems for mine detection under varied conditions [13].

2.11 Manipulators and Robotics Arm in Demining

Robotics manipulators are included in some robotics demining research to physically interact with or lift suspected objects without exposing humans to danger. The use of manipulators can enable safer verification and manipulation of potential threat [14].

2.12 Navigation, Autonomy and Control System

Autonomous navigation and path-planning are essential for robotics demining especially in complex or clustered environments. Autonomous re semi-autonomous system help to reduce operator load while maintaining safety and efficiency [15].

2.13 Real-World Deployments and Limitations

Despite progress, many robotics mine detection system are still at the prototype or research stage. Limitations include cost, environmental constraints (e.g. deep burial, soil heterogeneity), and robustness in real minefields [16].

2.14 Summery

The review literature that landmine detection is a highly dangerous and technically challenging task when performed using conventional manual methods. Most studies highlight robotics and intelligent systems as effective alternatives for operating in hazardous environment while minimizing human risk. Mobile robotics platform equipped with metal detector, ultrasonic and imaging sensor using GPS/GMS communication have shown improved efficiency in locating potential mines. The integration of robotics manipulators enables safe inspection and interaction with suspicious objects, while the application accuracy and reduces false alarms. However, many existing solutions remain limited by high cost, partial autonomy, environmental sensitivity and insufficient real-world validation. Therefore, there is a clear need for a low-cost, modular and intelligent robotics manipulator, which directly supports the motivation and objectives of this study.

3. Methodology

3.1 System Overview

The proposed system is an autonomous mobile robot integrated with a robotics manipulator and multiple sensors for landmine detection. It is controlled by an embedded microcontroller that coordinate sensing, motion and manipulation.

3.2 System Architecture (Block Diagram)



Fig. 3.1 Block Diagram

The microcontroller acts as the central unit. It receives input from the metal detector and ultrasonic sensors, processes the data, and sends control signals to the motor driver and servo motor. The Communication module allow remote monitoring and control, while the power supply provides regulated energy to all modules.

3.3 Hardware Design (Circuit Diagram)

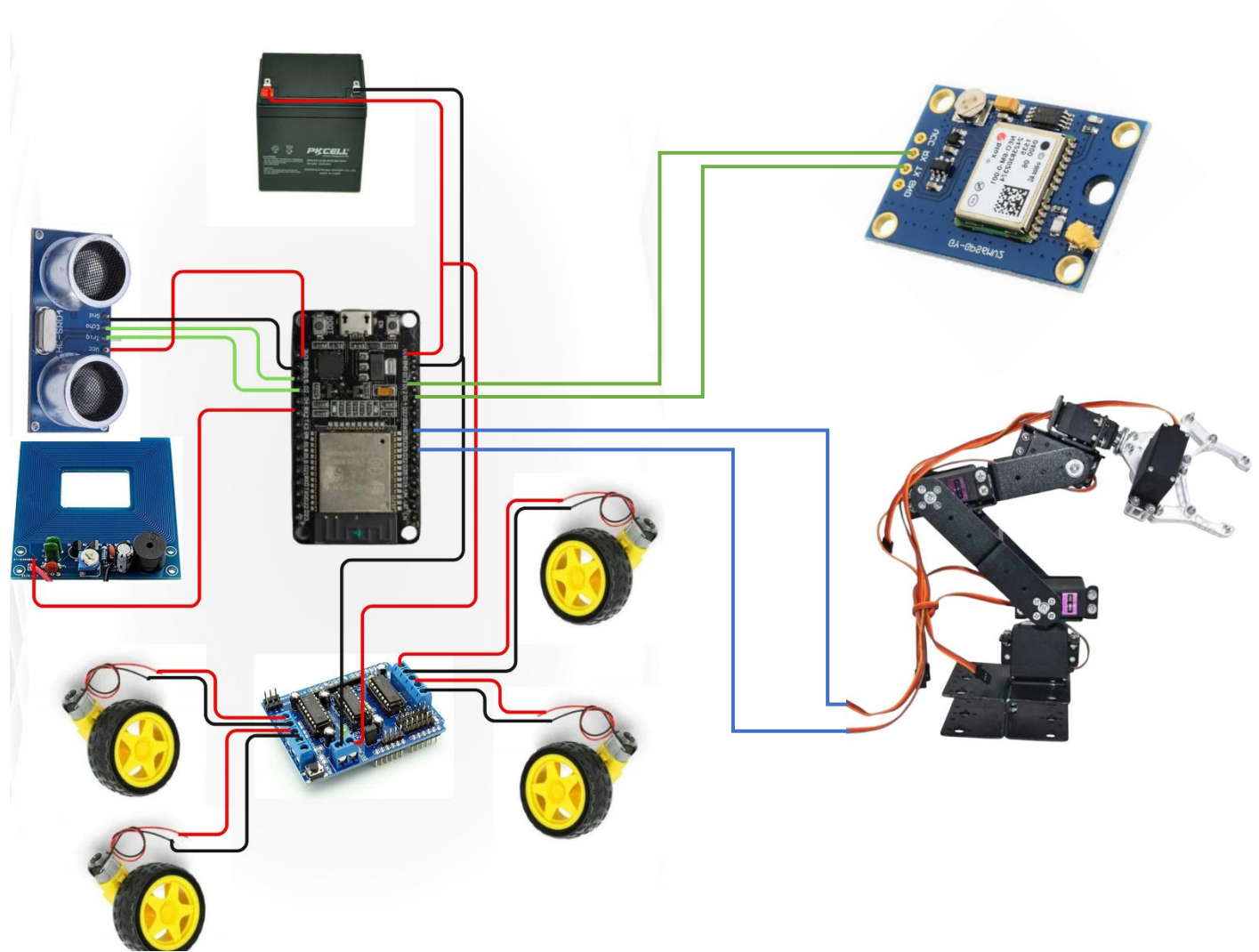


Fig. 3.2 Schematic Diagram

3.4 list of Components

1. ESP32 Microcontroller
2. Metal Detector Sensor
3. Ultrasonic sensor
4. DC Motor
5. Motor Driver
6. Servo Motor
7. Robotics Manipulator (Arm)
8. Mobile Robot Platform
9. Power Supply

3.4.1 ESP32 Microcontroller

The microcontroller acts as the central processing unit of the system. It continuously reads data from the sensors, processes this information using programmed logic and generates control signal for motors, servo actuators and communication module. The main purpose of using ESP32 is it coordinates all system operations including landmine detection, robot movement, obstacle avoidance and robotics manipulation control. Additionally, it has inbuilt Wi-Fi/Bluetooth module so we do not need to use additional wi-fi module.



Fig. 3.3 ESP32 Microcontroller

The ESP32 has following features and functionalities

- Dual-Core 32-bit microcontroller
- Operating frequency up to 24 MHz
- Built-in Wi-Fi (802.11 b/g/n) and Bluetooth (Classic +BLE)
- Multiple GPIO with ADC, DAC PWM support
- UART, SPI and I^2C communication interfaces
- Integrated timers, low-power and deep-sleep modes
- On-chip memory and flash support.

3.4.2 Metal Detector Sensor

The metal detector operates by generating an electromagnetic field. When a metallic object enters this field, it disturbs the signals which detected and converted into an electrical output signal



Fig. 3.4 Metal Detector Sensor

It is the primary sensor used to detect metallic components of buried landmine like objects. The main important features are;

- Detects metallic objects beneath the surface.
- Works using electromagnetic field sensing.
- Provide quick and reliable detection response.
- Can identify low and high metal content objects.
- Easy to interface with microcontroller (ESP32/Arduino).

3.4.3 Ultrasound Sensor

The ultrasound sensor emits high sound wave and measures the time taken for the echo to return after hitting an obstacle. This delay is used to calculate distance.



Fig. 3.5 Ultrasound Sensor

It is used for obstacle detection and safe navigation preventing collisions while the robot move in hazardous environments.

3.4.4 DC Motors

4 DC motors are used that convert energy into mechanical motion. This 4 DC motor used for wheel of the car to move in the area. Their speed and direction are controlled by varying the voltage and magnitude through a motor driver.

They provide movement to the mobile robotics platform mean the car which move in the certain are for scanning and it enable area coverage for landmine detection.



Fig. 3.6 DC motors for Wheel

3.4.5 Motor Driver (L293D)

The Motor driver acts as an interface between the low-power microcontroller and high-power DC motors. It amplifies control signals and allows bidirectional motor rotation.

It enables safe and efficient control of the robot movement without overloading the microcontroller.



Fig 3.7 L293D Motor Driver

3.4.6 Servo Motors

The Servo motors is used for the movement of manipulator (Arm). It is 6 DOF and operate using pulse width Modulation (PWM) signal to achieve precise positioning

They control the joint of the robotics manipulator, allowing accurate movement for inspection and handling of detected objects.



Fig. 3.8 Servo Motor

3.4.7 Robotics Manipulator

A multi-DOF (6 DOF) robotics arm driven by servo motors is used to inspect and handle detected objects without human contract. The Arm is metallic and can carry 1 kilogram weight. When the metal detector detect object the microcontroller send signal to the arm based on the signal received from metal detector and the arm pick the landmine or the metal object from ground.



Fig. 3.9 Robotics Manipulator

The Manipulator has following important features and functionalities;

- 6-DOF (base rotation, shoulder, elbow, wrist pitch, wrist yaw and wrist roll)
- High positional and orientation flexibility
- Servo-motor based precise joint control
- Capable of complex 3D movement
- Programable motion and repeatable accuracy

- Grasping and lifting detected objects safely
- Precise positioning near suspicious landmine-like object
- Remote and automated operation through ESP32
- Minimize human involvement in hazardous handling

3.4.8 Mobile Robot Platform

A Wheeled chassis with 4 DC motors is used to move the robot over the test area. This body is made of plastic material. The motors is connected to the motor driver which enable the motors to move in the area while providing necessary power. The slow and controlled movement ensure accurate sensing and operational safety

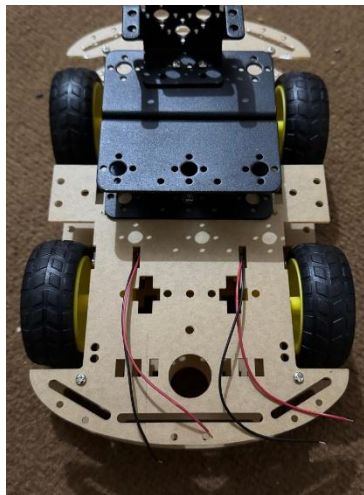


Fig. 3.10 Robotics Car Platform

3.4.9 Power Supply Unit and Voltage Regulator

A rechargeable battery supplies DC power to the system. Voltage regulators ensure stable voltage levels for sensitive components. It provides portable and uninterrupted power for autonomous

operation. The voltage regulator maintains a constant output voltage despite variations in input or load. It protects electronic components from voltage fluctuations and ensures reliable operation.



Fig. 3.11 Power Supply (Battery)

3.5 Software and Control Algorithm

The microcontroller (ESP32/Arduino) is programmed with embedded software to read sensor inputs, run control logic continuously, and make decisions in real time. It computes the robot's movement commands, checks ultrasonic sensors for obstacles, and keeps an eye on the metal detector for objects that resemble landmines. Additionally, the algorithm produces Pulse Width Modulation (PWM) signals to precisely position the manipulator joints and regulate the speed and direction of DC motors. It also manages Bluetooth or Wi-Fi communication with the operator, enabling remote system monitoring and control.

It allows for Intelligent behavior, making sure that when a possible mine is found, the robot stops right away and turns on the manipulator for a secure inspection. All components are seamlessly

coordinated by the software, enabling precise object manipulation, easy navigation, and obstacle avoidance. All things considered, it increases operational safety, increases the dependability and effectiveness of landmine detection, and gives the robot some degree of autonomy.

3.6 Mechanical Structure

Operating: The chassis ensures stability and alignment by supporting all mechanical and electronic components. It offers a stiff platform on which the robotic manipulator, microcontroller, sensors, and motors can be mounted. Accurate operation of sensors and actuators is ensured by the mechanical structure's ability to absorb small shocks and vibrations during movement. Additionally, it keeps the manipulator from toppling over when it is extended by maintaining appropriate weight distribution. To enable smooth and controlled motion throughout the test environment, wheels and motor mounts are integrated into the chassis.

Purpose: Its role in the project is to maintain structural integrity and facilitate easy mobility throughout the testing environment. By keeping all parts securely in place and facilitating dependable navigation over uneven surfaces, it also guarantees safe operation.

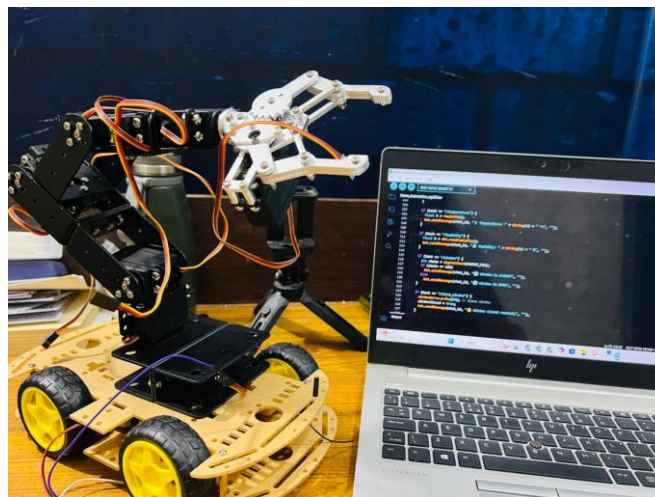


Fig. 3.12 Mechanical Structure

3.7 Material and Development Cost

The main hardware elements in the creation of the robotic landmine detection system are shown in the following table along with an estimate of their market value. The suggested system is economically viable and appropriate for low-cost humanitarian applications, as this cost analysis shows.

Sr. No.	Component Name	Quantity	Unit Cost (PKR)	Total Cost (PKR)
1	ESP32 Microcontroller	2	2,000	2,000
2	Metal Detector Sensor	1	550	550
3	DC Gear Motors	4	200	800
4	Motor Driver (L298N)	1	500	500
5	6-DOF Robotic Arm	1	12,000	12,000
6	Ultrasonic Sensor	1	200	200
7	Chassis & Wheels	1	1500	1500
8	Power Supply & Batteries	1	2,500	2,500
9	Connecting Wires & Misc.	1	500	500
10	Shipment Charges	1050		
11	Insurance	300		
12	Total	21000/-		

Table. 1 Estimate Cost from bingeing to present

3.8 Summery

The design, integration, and operation of an intelligent robotic system for secure landmine detection are the main objectives of this project's methodology. The system is made up of a mobile robotic platform with wireless communication modules, an ESP32/Arduino microcontroller, metal detection and ultrasonic sensors, and a 6-DOF manipulator. Each part is carefully chosen to carry out a particular task: the manipulator enables accurate inspection and handling of suspicious objects, the motors provide controlled mobility, the microcontroller coordinates sensing and actuation, and the sensors identify metallic objects and obstacles. In order to process sensor data, navigate safely, and react intelligently to threats, the embedded software runs real-time control algorithms. While the power supply ensures continuous operation, the chassis and mechanical structure guarantee stability and dependability during movement. In order to ensure that the robotic system can safely and effectively detect, approach, and manipulate potential landmine objects in a controlled environment, the methodology generally places an emphasis on modular design, component coordination, and semi-autonomous operation

4. Result

The developed robotics system successfully demonstrates the core functionality of autonomous metal detection and mechanical response. During experimental testing, the metal detector sensor consistently identified the presence of metallic object within its effective range, and upon detection, the ESP32-based control unit immediately activated the 6-DOF manipulator arm. The arm performed the programmed motion sequence smoothly and reliably, conforming proper coordination between sensing, control and actuation modules. This achievement validates the feasibility of the proposed design for preliminary landmine detection applications and establishes a strong foundation for integrating advanced navigation, communication and intelligent decision-making features in future stages of the project.

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